

CUSTOMER MILESTONE BRIEF

MS #37 - PL/I Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.37.0
Date	2026-08-30
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #37 - PL/I Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Expands and governs PL/I evidence across language semantics and mainframe integration boundaries.

Why this matters to a customer

Expands and governs PL/I evidence across language semantics and mainframe integration boundaries.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Consolidated the PL/I conformance lab, mixed-language development proof, mutation evidence, and candidate build attestation into ten independent qualification gates.
- Added a 20-entry PL/I compatibility ledger covering declarations, numeric and character semantics, storage, preprocessing, calls, conditions, files, Db2, CICS, IMS, and IBM runtime boundaries using all five governing classifications.
- Expanded the conformance corpus from 27 to 52 cases with 25 targeted boundaries for storage classes, arrays, pointers, picture/decimal arithmetic, conditions, preprocessing, record layouts, parameter conventions, SQL cursors, CICS, and IMS; unsupported behavior fails closed.
- Bound the qualification to 52 synthetic cases, 22 supported construct categories, seven mutation cases, 16 blocked cases, and the bounded ACCTPL1 reference workload.

- Added deterministic qualification artifacts, frozen schemas, Linux/Windows wrappers, release verification integration, and fail-closed overclaim tests.
- Kept general Enterprise PL/I, native compiler, z/OS runtime, mainframe equivalence, and production readiness false; the corpus contains no customer source.

How it differs from earlier work

MS #37 builds on MS #36 (COBOL Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds pl/i qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use pl/i qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept general Enterprise PL/I, native compiler, z/OS runtime, mainframe equivalence, and production readiness false; the corpus contains no customer source.

Source of record

- CHANGELOG.md release section(s): 0.37.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.